

Tahsin Reza

[Website](#) | [Email](#) | [LinkedIn](#) | [GitHub](#) | [ORCID](#)

EDUCATION

Bangladesh University of Engineering and Technology (BUET), Dhaka

Jan 2022 - June 2026

Bachelor of Urban and Regional Planning (BURP)

CGPA: 3.70/4.00

Relevant Coursework: Statistics I and II, Programming Techniques, GIS and Remote Sensing, Transportation Policy and Planning, Environmental Planning and Urban Planning Techniques

RESEARCH INTEREST

Travel Behavior & Mode Choice, Traffic Modeling and Micro-simulation, Data Science for Transport Planning, Geospatial Analytics for Transportation, Urban Mobility, Safety, Accessibility & Equity in Transportation

ACADEMIC HONORS AND AWARDS

- University Merit Scholarship, BUET (2023)

RESEARCH EXPERIENCE

[1] Time-Headway Analysis of Pedal Rickshaws and Battery-Operated Three-Wheelers in Dhaka's Non-Lane-Based Heterogeneous Traffic

- Built an automated extraction pipeline, a YOLO26 detector trained on roughly 207,000 self-labeled vehicle instances paired with BoT-SORT tracking, producing about 900 follower-leader observations across two corridors that feed both manuscripts below

(a) Pair-Specific Time Headway Distributions and Covariate-Conditioned AFT Models for Pedal Rickshaws and Battery-Operated Three-Wheelers in Dhaka's Non-Lane-Based Heterogeneous Traffic

Manuscript Submitted (Transportation Research Board (TRB) Annual Meeting 2027, 1st Author) [\[Online Version\]](#)

- Fitted pair-specific headway distributions and accelerated failure time (AFT) models to explain distributional characteristics and shifts, accounting for covariate effects that feed both simulation modeling and the policy horizon.

(b) One Label, Two Vehicles: Time-Headway Evidence Against Treating Pedal and Battery Rickshaws Alike in Mixed Traffic

Manuscript in Preparation

- Battery rickshaws followed 30–39% closer than pedal rickshaws; the gap ran overwhelmingly through speed (66–80%), with physical spacing contributing modestly (10–11%). Off-centered following independently shortened headway by a replicated 10–12%. Exposure to critically short following (under 2.0 s) was 2.5 times higher for battery rickshaws, rising to 3.8–4.5 times at 1.5 s.

[2] Mono- versus Intermodal Trip Choice in Dhaka: Bayesian Multilevel Models and Explainable Machine Learning

Manuscript in Preparation

- Built a four-level (M0–M3) Bayesian multilevel logistic regression and an explainable Random Forest/SHAP pipeline to model monomodal vs. intermodal trip choice across 25,242 trips nested within 3,729 Dhaka households.
- Confirmed intermodality is necessity-driven, not discretionary: households below Tk 40,000/month income and in tin-shed housing (OR $\approx$ 5.9 vs. self-built housing) chained modes most, while most vehicle-ownership categories cut the odds by $\sim$ 90% and higher income by $\sim$ 75%, the reverse of the affluence-linked intermodality reported in European studies.

[3] Flow Characteristics of Traditional Rickshaws and Battery-Operated Three-Wheelers in Non-Lane-Based Heterogeneous Traffic of Dhaka

Bachelor's Thesis, Department of Urban and Regional Planning, BUET (2026) [\[Thesis Report\]](#)

- Conducted videographic surveys at three Dhaka corridors, manually extracting 531 valid target-leader time-headway observations using a reference-line method.
- Benchmarked six tree-based models and selected a tuned CatBoost model (Test $R^2 = 0.674$, MAE = 0.400 s) with SHAP analysis identifying speed as the strongest predictor, confirming battery-operated three-wheelers consistently follow with shorter headways than rickshaws - evidence against regulating them as one vehicle class.

SELECTED UNDERGRADUATE PROJECTS

[1] Analyzing Core Rural Road Networks and Land Use Conversion: A Case Study on Selected Upazilas in Bangladesh

Rural Planning Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed all unions across 14 upazilas via KIIs and GIS buffer analysis (LGED roads, BBS demographics, Sentinel land-use), then built a log-log regression model ($R^2=0.44$, $p<0.001$) of core road density. Population ($\beta=0.305$) and education-institution ($\beta=0.205$) density were strongest predictors. Core roads served ~90% of growth centers/markets, 50–60% of key facilities, amid 81% natural land-use conversion.

[2] A Study on Factors Influencing Access Mode Choice of Metro Rail (MRT) Users in Dhaka City

Social and Physical Survey Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed 345 MRT users at 3 stations to identify first/last-mile access mode determinants. Rickshaw dominated (41%), walking/bus tied at 20%; 84%+ traveled under 5 km and about 77% of trips under 1 km were made on foot. Walking led in Pallabi (shortest distance), rickshaw in Uttara North (lower crime risk), and bus in Agargaon (consistent road width).

[3] Factors Influencing Pedestrian Catchment Area of Major MRT Stations in Dhaka

Transport Planning Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed 717 commuters at Motijheel and Farmgate using random sampling and station-gate video counts, mapping built-environment variables across six GIS buffer rings (250–2000 m) and training a Random Forest Regressor to predict egress-walk distance. Mean distances were 440.8 m and 401.7 m, with 12–17% exceeding the conventional 800 m catchment radius; the model achieved $R^2 = 0.94$, with road density and floor area outweighing sociodemographic factors.

[4] Transportation Problem Solver: North-West Corner and Least Cost Methods [\[Presentation\]](#)

Programming Techniques, Department of Urban and Regional Planning, BUET

- Built a Python tool that computes and compares initial feasible solutions to the classical transportation problem using both the North-West Corner and Least Cost methods.

PROFESSIONAL EXPERIENCE

Assistant Urban Planner

July 2026 - Present

Sheltech Consultants (Pvt.) Ltd., Dhaka

- Developed the transport planning component of a Pourashava master plan report, introducing capacity, flow, level-of-service, and Space Syntax analyses; based on the survey report and ToR, formulated the report's volume structure.
- Supported fellow planners across the wider project team in implementing the methodology and resolving transport-related planning questions for the remaining pourashavas.

Intern

November 2025 - December 2025

Dhaka Transport Coordination Authority (DTCA), Dhaka

- Assessed the traffic impact of converting a road in Dhaka into a pedestrian precinct. Ran traffic and pedestrian surveys, then built and compared before-and-after microsimulation scenarios in PTV Vissim to test the proposed layout.

TEACHING EXPERIENCE

Tutoring Secondary and Higher Secondary students (Physics, Math, ICT, Chemistry)

2021-Present

- Provided academic support to over 12 students across varying levels
- Served as a Mathematics instructor at a local coaching center.

SOFTWARE AND PROGRAMMING SKILLS

Programming Language: Python, R, C | **GIS Tool:** ArcGIS, ArcGIS Pro, Google Earth Engine | **Transportation Modelling:** PTV Vissim | **Modeling and ML:** Discrete Choice Models, Survival/AFT Models, Tree-Based Ensemble Models, SHAP, YOLO Object Detection, scikit-learn, Distribution Fitting (scipy) | **Design and Drafting:** AutoCAD, Adobe Illustrator, Adobe Photoshop | **Productivity:** MS Word, Excel, PowerPoint, Project, CapCut

EXTRA-CURRICULAR ACTIVITIES

- Secretary, Student Chapter, **Bangladesh Institute of Planners (BIP)** (2025-2026)
- Vice President, **URP Students' Association of BUET (USAB)** (2025-2026)
- Joint General Secretary, **Help_BUETian** (2025-2026)

REFERENCES

Dr. Md. Musleh Uddin Hasan

Professor
Dept. of Urban and Regional Planning
Bangladesh University of Engineering and Technology
Email: musleh_uddin@urp.buet.ac.bd
Relationship: Thesis Supervisor and Research Mentor

Ms. Sadia Afroj

Assistant Professor
Dept. of Urban and Regional Planning
Bangladesh University of Engineering and Technology
Email: afrojsadia@urp.buet.ac.bd
Relationship: Course Instructor and Research Mentor

Dr. Md. Asif Raihan

Professor, Accident Research Institute (ARI)
Associate Director, BUET Career Center (BCC)
Bangladesh University of Engineering and Technology
Email: raihan@ari.buet.ac.bd; araihan002@yahoo.com
Cell: +8801911142802
Relationship: Research Mentor